

Project: Aura Retail OS (Sub Task – 1)

Designing an Autonomous Modular Smart-City Retail Infrastructure

Team Members:

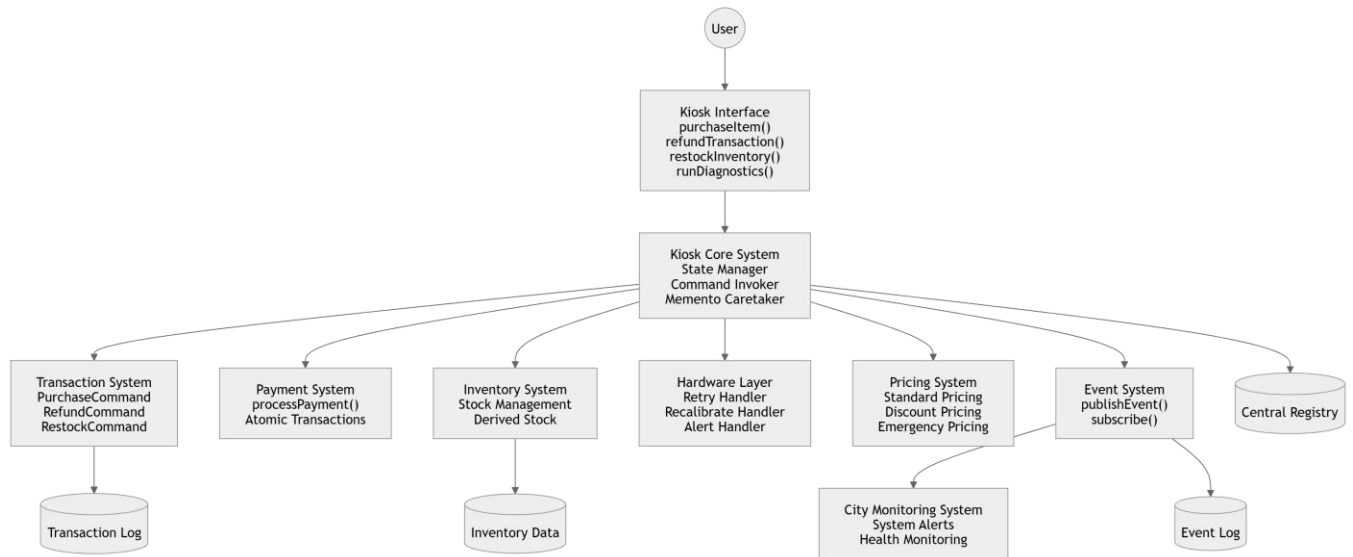
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1) High-level architecture diagram:



2) Preliminary class diagram:



3) Planned Design Patterns

Pattern	Where Used	Why
Singleton	Central Registry	Maintains a single global source of truth for configuration and system state
Facade	Kiosk Interface	Provides a simple unified interface for all user operations
Abstract Factory	Kiosk Creation (Pharmacy, Food, Emergency)	Creates environment-specific components without changing core logic
Command	Transaction Manager (Purchase, Refund, Restock)	Encapsulates operations for logging, undo, and consistency
Observer	Event Bus	Enables event-driven communication between loosely coupled subsystems
Strategy	Policy Manager (Pricing System)	Allows dynamic switching of pricing strategies at runtime
State	Kiosk Mode Manager	Changes system behavior based on operational mode
Chain of Responsibility	Failure Handler Pipeline	Processes failures step-by-step (retry → recalibrate → alert)
Memento	Transaction Snapshot System	Saves system state for rollback during failures
Mediator	Decision Engine	Coordinates between pricing, modes, and transactions without tight coupling

4) Subsystem Responsibilities of Each Member

Member 1 (Jay) – Kiosk Core System (Integration Layer)

Responsibilities:

- Acts as the central controller of the system.
- Handles all user requests coming from the Kiosk Interface.
- Coordinates communication between Inventory, Payment, and Hardware systems.
- Maintains low coupling between subsystems using interfaces.

- Controls overall system flow and decision-making.

Key Functionalities:

- Processes operations such as purchase, refund, and restock.
- Manages the transaction flow:
 - Receive request
 - Check inventory
 - Process payment
 - Trigger hardware
- Handles success and failure cases.
- Manages kiosk operational modes such as Active and Emergency.

Design Responsibilities:

- Implements Facade Pattern through KioskInterface.
- Uses State Pattern for managing kiosk modes.
- Coordinates execution of commands.
- Interacts with CentralRegistry for system state.

Member 2 (Priya) – Inventory & Product Management System**Responsibilities:**

- Manages product details, stock levels, and inventory data.
- Ensures inventory consistency throughout the system.
- Maintains accurate stock during concurrent transactions.

Key Functionalities:

- Checks product availability before purchase.
- Reserves items during transactions.
- Updates stock after successful transactions.
- Computes available stock dynamically based on:
 - total stock
 - reserved items
 - faulty items
- Enforces constraints such as:
 - no purchase when stock is zero

- limits during emergency mode

Additional Duties:

- Triggers LowStockEvent when stock falls below threshold.
- Marks items unavailable if required hardware is not available.
- Supports extension for product bundles.

Design Responsibilities:

- Maintains consistency and correctness of inventory.
- Works with Event System for notifications.

Member 3 (Priya) – Payment & Transaction Management System

Responsibilities:

- Handles all financial transactions in the system.
- Ensures secure and reliable payment processing.
- Maintains transaction history and logs.

Key Functionalities:

- Processes payments using multiple methods such as UPI, card, and wallet.
- Validates transactions before processing.
- Handles payment success and failure scenarios.
- Maintains transaction records.

Constraints Handling:

- Ensures atomic transactions:
 - complete fully or rollback completely
- Performs rollback in case of failure.

Additional Duties:

- Coordinates with Inventory System for stock reservation.
- Prevents transactions if hardware dependency fails.
- Handles delayed hardware response safely.

Design Responsibilities:

- Implements Command Pattern for operations.
- Uses Memento Pattern for rollback.
- Uses Strategy Pattern for different payment methods.

Member 4 (Jenil) – Hardware, Event & Monitoring System

Responsibilities:

- Controls hardware components such as motors, sensors, and dispensers.
- Provides abstraction between software and hardware.
- Manages system events and external notifications.

Hardware Functionalities:

- Executes product dispensing.
- Monitors hardware health and status.
- Performs system diagnostics.
- Handles hardware delays and failures.

Event & Monitoring:

- Implements event-driven communication.
- Handles events such as:
 - LowStockEvent
 - HardwareFailureEvent
 - EmergencyModeActivated
- Sends alerts to city monitoring and maintenance systems.

Failure Handling:

- Implements recovery mechanism:
 - Retry
 - Recalibration
 - Technician alert

Additional Responsibilities:

- Ensures system stability during concurrent operations.
- Updates system state in CentralRegistry.
- Communicates with external systems.

Design Responsibilities:

- Uses Observer Pattern for event handling.
- Uses Chain of Responsibility for failure handling.
- Implements hardware abstraction layer.